

~~SECRET~~

IT/IG/P&C Branch/Collection Sect
Jol Ligon/2931/ew

DEPARTMENT OF THE ARMY
General Staff, United States Army
Intelligence Division
Washington 25, D. C.

INTELLIGENCE COLLECTION)

21 January 1948

MEMORANDUM NUMBER 7)

UNCONVENTIONAL AIRCRAFT

OBJECT

1. The object of this memorandum is to enunciate partial and continuing requirements for information in the field of unconventional aircraft.

GENERAL

2. The German High Command indicated a definite interest in the Horten type of flying wing and were about to embark on a rigorous campaign to develop such aircraft toward the end of the war. A Horten design known as the IX which was designated as the Go-8-229 and Go-P-60 (night fighter) was to be manufactured by the Gotha Plant. It is reported that a contract for fifty such aircraft was planned but only three or four were built. This plant is now in the hands of the Russians. A recent report indicates that the Russians are now planning to build a fleet of 1800 Horten VIII (six engine pusher) type flying wing aircraft. The wing span is 131 feet. The sweep-back angle is 30 degrees. The Russian version is reported to be jet propelled.

REQUIREMENTS

3. Requirements appear at inclosure.

SPECIAL INSTRUCTIONS

4. The inclosure was prepared to insure collection action principally in the field of Soviet activities in the U.S.S.R. Portions, however, have equal application to Soviet activities outside the U.S.S.R. and to activities of non-Soviet agencies and individuals located within areas of responsibility of addressees. The inclosure will therefore be applied locally both as an enunciation of specific requirements and as a general guide for collecting and reporting in the subject field.

DECLASSIFIED

NND 755004

By 7 NARS, Date 1/10/79

34080

~~SECRET~~

RECORD SECTION COPY

SECRET

CONTROL

5. Replies to the specific requirements at inclosure will refer to Control Number A-1917. Replies to other unconventional aircraft requirements will refer to other appropriate control numbers, if any.

FOR THE DIRECTOR OF INTELLIGENCE:

MAILED 12, WOODS 21 JAN 1957

R. F. Ennis
R. F. ENNIS
Colonel, GSC
Chief, Intelligence Group

Incl:

Intelligence Requirements -
Unconventional Aircraft

DISTRIBUTION

For Action

Commander-in-Chief, Far East, APO 500, c/o Postmaster, San Francisco, California (3 copies)
Commanding General, U. S. Army Forces in Korea, APO 235, c/o Postmaster, San Francisco, California
Commander-in-Chief, European Command, ATTENTION: A. C. of S., G-2, APO 757, c/o Postmaster, New York, New York (2 copies)
Commanding General, U. S. Forces, Alaska, APO 942, c/o Postmaster Seattle, Washington
Commanding General, U. S. Forces, Austria, APO 777, c/o Postmaster, New York, New York
Commanding General, Trieste, United States Troops, APO 290, c/o Postmaster, New York, New York
Military Attache, American Embassy, Brussels, Belgium
Military Attache, American Legation, Sofia, Bulgaria
Military Attache, American Embassy, Ottawa, Canada
Military Attache, American Embassy, Nanking, China
Military Attache, American Embassy, Praha, Czechoslovakia
Military Attache, American Embassy, Copenhagen, Denmark
Military Attache, American Legation, Helsinki, Finland
Military Attache, American Embassy, Paris, France
Military Attache, American Embassy, London, England
Military Attache, American Embassy, Athens, Greece
Military Attache, American Legation, Budapest, Hungary
Military Attache, American Embassy, Teheran, Iran

(Cont'd on page 3)

SECRET

SECRET

For Action Cont

Military Attache, American Embassy, Baghdad, Iraq
Military Attache, American Embassy, Rome, Italy
Military Attache, American Embassy, The Hague, The Netherlands
Military Attache, American Embassy, Oslo, Norway
Military Attache, American Embassy, Warsaw, Poland
Military Attache, American Legation, Bucharest, Rumania
Military Attache, American Embassy, Madrid, Spain
Military Attache, American Legation, Stockholm, Sweden
Military Attache, American Legation, Bern, Switzerland
Military Attache, American Embassy, Ankara, Turkey
Military Attache, American Embassy, Moscow, U.S.S.R.
Military Attache, American Embassy, Belgrade, Yugoslavia

For Information

Special Asst. for Research and Intelligence, Department of State
Central Intelligence Agency
Chief of Naval Intelligence, Department of the Navy
Director of Intelligence, Department of the Air Force (6 copies)

-3-
SECRET

~~SECRET~~

INTELLIGENCE REQUIREMENTS - UNCONVENTIONAL AIRCRAFT

1. Research and Development

- a. What German scientists had a better-than-average knowledge of the Horten brothers' work and perspective thinking; where are these scientists now located, and what is their present activity? Should be contacted and interrogated.
- b. What Russian factories are building the Horten VIII design?
- c. Why are the Russians building 1,800 of the Horten VIII design?
- d. What is their contemplated tactical purpose?
- e. What is the present activity of the Horten brothers, Walter and Riemer?
- f. What is known of the whereabouts of the entire Horten family, particularly the sister? All should be contacted and interrogated regarding any contemplated plans or perspective thinking of the Horten brothers, and any interest shown by the Russians to develop their aircraft.
- g. Are any efforts being made to develop the Horton "Parabola" or modify this configuration to approximate an oval or disc?
- h. What is the Horton perspective thinking on internal controls or controls that are effective mainly by streams of air or gas originating from within the aircraft to supplant conventional external surface controls?

2. Control

For any aircraft whose shape approximates that of an oval, disc, or saucer, information regarding the following items is requested:

- a. Boundary layer control method by suction, blowing, or a combination of both.
- b. Special controls for effective maneuverability at very slow speeds or extremely high altitudes.
- c. Openings either in the leading edge top and bottom surfaces that are employed chiefly to accomplish boundary layer control or for the purpose of reducing the induced drag. Any openings in the leading edge should be reported and described as to shape, size, etc. This investigation is significant to justify a disc shape configuration for long-range application.

~~SECRET~~

SECRET

- d. Approximate airfoil shape in the center and near the tips.
- e. Front view and rear view shape.

3. Items of Construction

- a. Type of material, whether metal, ferrous, non-ferrous, or non-metallic.
- b. Composite or sandwich construction utilizing various combinations of metals, plastics, and perhaps balsa wood.
- c. Unusual fabrication methods to achieve extreme light weight and structural stability particularly in connection with great capacity for fuel storage.

4. Items of Arrangement

- a. Special provisions such as retractable domes to provide unusual observation for the pilot or crew members.
- b. Crew number and accommodation facilities.
- c. Pressurized cabin equipment.
- d. High altitude or high speed escapement methods.
- e. Methods of pressurization or supercharging from auxiliary units or from the prime power plant.
- f. Provisions for towing - especially with short fixed bar, and for re-fueling in flight.
- g. Provisions for assisted take off application.
- h. Bomb bay provisions, such as dimensions, approximate location, and unusual features regarding the opening and closing of the doors.

5. Landing Gear

- a. Indicate type of landing gear - whether conventional, tricycle, multiple wheel, etc.
- b. Retractable, and jettison features for hand gear.
- c. Provisions for take off from ice, snow, or water.
- d. Skid arrangements for either take off or landing.

SECRET

SECRET

6. Power Plant

a. Information is needed regarding the propulsion system used in the aircraft. Possible types of engines that could be employed include:

- (1) Reciprocating (piston type) engine or gas turbine. Either or both of these could be used to drive propellers of conventional or special design, rotating vanes, ducted fans, or compressors.
- (2) Jet propulsion engines including turbo jets, rockets, ram-jets, pulse jets, or a combination of all four.
- (3) Nuclear propulsion (atomic energy). Atomic energy engines would probably be unlike any familiar type of engine, although atomic energy might be employed in combination with any of the above types.

Aircraft would be characterized by lack of fuel systems and fuel storage place.

b. The power plant would likely be an integral part of the aircraft and could possibly not be distinguished as an item separate from the aircraft. If jet propulsion is used, large air handling capacity, characterized by a large air inlet and large exhaust nozzle, should be evident. The size of entrance and exit areas would be of interest. It is possible that the propulsive jet is governed or influenced for control of the aircraft. The presence of vanes or control surfaces in the exhaust or methods of changing the direction of the jet should be observed.

c. Information desired on the propulsion systems pertains to the following items:

- (1) Type of power plant or power plants.
- (2) General description.
- (3) Rating (thrust, horsepower, or air flow).
- (4) Type of fuel.
- (5) Catalytic agents for super-performance or normal cruising power.